

SIMBARASHE MTHANDAZO MPOFU

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PROFILE

Biotechnology graduate with hands-on research experience in environmental health, antimicrobial resistance, molecular biology and scientific data analysis. Experienced in conducting literature reviews, collecting and analysing experimental data, evaluating evidence, preparing scientific reports and interpreting findings against published research and established standards. Complementary professional experience in laboratory quality assurance, documentation, auditing and technical communication. Detail-oriented and rigorous in evidence evaluation, fact-checking and scientific reporting, with a strong interest in research addressing the intersection of environmental change, health and human wellbeing.

EDUCATION

2020-2024	National University of Science and Technology BSc. Hon. Biotechnology
2015-2016	Rimuka 2 High School ZIMSEC GCE Advanced Level
2010-2013	Sizane High School ZIMSEC GCE Ordinary Level
2012-2013	Sizane High School National Foundation Certificate in Computer Operations and Packages

RESEARCH EXPERIENCE

Undergraduate Research Project — BSc Honours Biotechnology

National University of Science and Technology | 2024

Project Title: A Molecular Survey of Antifungal Susceptibility Patterns and a Selected Virulence Factor of *Candida albicans* from Mzingwane and Mtshabezi Dams

- Designed and conducted a four-month research investigation examining antifungal susceptibility patterns and molecular markers associated with antifungal resistance and virulence in environmental *Candida albicans* isolates.
- Collected composite environmental water samples from multiple sites at Mzingwane and Mtshabezi Dams and isolated and confirmed *C. albicans* using microbiological and biochemical methods.
- Conducted antimicrobial susceptibility testing using the Kirby-Bauer disc diffusion method and evaluated resistance patterns against fluconazole, itraconazole and posaconazole.
- Performed DNA extraction and conventional PCR to investigate **ERG11, MDR1 and ALS1** genes associated with antifungal resistance and virulence.
- Analysed and interpreted experimental findings, including susceptibility percentages and PCR results, and evaluated findings against published scientific literature.
- Conducted an extensive literature review and critically discussed possible environmental and clinical factors contributing to antifungal resistance.
- Produced a formal scientific dissertation documenting the research methodology, results, discussion, conclusions and recommendations.

RESEARCH AND ANALYTICAL COMPETENCES

Research

Literature review • Evidence synthesis • Scientific research methodology • Environmental sampling • Experimental design

Analysis

Quantitative data analysis • Data interpretation • Antimicrobial susceptibility analysis • Molecular data interpretation • Critical evaluation of evidence

Scientific methods

Microbial isolation and identification • Kirby-Bauer disc diffusion • DNA extraction • PCR • Agarose gel electrophoresis • PCR optimisation

Writing & communication

Scientific writing • Technical reporting • Research documentation • Literature-based discussion • Presentation of research findings

Quality

Quality assurance • Data verification • Fact-checking • Accurate record keeping • Scientific protocol compliance

WORKING EXPERIENCE

LABORATORY TECHNICIAN

Higher Learning Centre, Bulawayo, Zimbabwe | 2025-present

- Maintain accurate laboratory records, documentation and inventory data to support effective resource management and accountability.
- Conduct systematic stock audits by comparing physical inventory against departmental records, identifying discrepancies and documenting findings for management review.
- Analyse inventory and resource requirements to identify shortages, forecast replenishment needs and support departmental budgeting and procurement planning.
- Prepare technical documentation, reports and administrative records relating to laboratory operations, resources and quality assurance.
- Apply established quality, safety and procedural standards to laboratory activities, equipment and materials
- Prepare reagents, experimental materials and practical workstations according to specified procedures and requirements.
- Support scientific practical activities by coordinating materials, equipment and workstations and communicating technical information clearly to teachers and students.
- Contribute to continuous improvement of laboratory systems, organisation and operational efficiency.

LABORATORY TECHNICIAN/SCIENTIST

Immunogenetics Laboratories, Bulawayo, Zimbabwe | 2022-2023

- Conducted laboratory analyses according to established scientific procedures and quality-assurance protocols.
 - Collected, recorded and maintained accurate experimental and laboratory data to support reliable results.
 - Maintained comprehensive laboratory documentation and records, ensuring traceability and accuracy of information.
 - Applied quality-control procedures to laboratory processes and identified procedural issues requiring investigation or corrective action.
 - Assisted with troubleshooting laboratory procedures and contributed to maintaining efficient analytical workflows.
 - Operated laboratory equipment in accordance with established procedures and maintained readiness for analytical work.
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REFERENCES

Dr. Edward Zumbika
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